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Effect of Multiple Representations to Promote Early Algebraic Thinking in Students with Autism

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ABSTRACT

Given the growing emphasis on algebraic concepts in primary education curricula and the importance to investigate how students with Autism Spectrum Disorder (ASD) approach mathematical learning, the purpose of this study is to evaluate the effectiveness of an instruction based on multiple representations in promoting early algebraic thinking in students with ASD. Three students with ASD, aged 7, 7, and 9 years, enrolled in the same general education school in Spain participated in the study. The research followed a single-subject, multiple probe across participants design. The results of the study showed that the participants improved their performance on generalization tasks in functional contexts when using multiple representations. These improvements were maintained over time and transferred to everyday situations involving regularities. Social validity data collected from students' families and teachers indicated that both groups had positive perceptions of students' attitudes and motivation after the study. The implications of the results for promoting early algebraic thinking in students with ASD are discussed.

Introduction

In general, students with Autism Spectrum Disorder (ASD) tend to have lower math performance than students without ASD (Estes et al., 2011). While some students with ASD may maintain satisfactory math performance during the early years of primary school, they frequently encounter challenges in later grades, particularly at the beginning of secondary school when abstract reasoning becomes crucial (Cleary & Hart Barnett, 2015). Studies examining mathematical learning trajectories for students with disabilities have shown that the hierarchical nature of mathematics learning (Xu et al., 2023) often leads to the accumulation of gaps in students' foundational knowledge, impeding their progress in grasping new concepts (Wei et al., 2013). In particular, students with ASD show similar slopes and accelerations in applied problems compared to other disabilities, but slower progress across grades in calculation problems (Wei et al., 2013).

The aforementioned challenges stem, among other factors, from difficulties in skills related to executive functions and communication (Polo-Blanco et al., 2024; Root & Browder, 2019). As a consequence, a substantial number of students with ASD require educational support to facilitate their learning of mathematics. In particular, these difficulties can negatively affect different areas such as number-word sequence, calculation, fact retrieval, and problem-solving (Bae et al., 2015; Donlan, 2007; Polo-Blanco et al., 2024). Therefore, it is important to investigate how students with ASD approach mathematical tasks, in order to develop customized instructional approaches that enable them to make progress according to their individual abilities.

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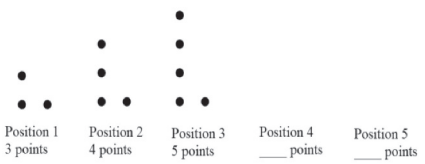
Most research on mathematical learning in students with ASD has focused on numerical aspects and arithmetic problem-solving in primary school (e.g., García-Moya et al., 2023; Goñi-Cervera et al.2024; Polo-Blanco & González, 2021; Rockwell et al., 2011; Root et al., 2017), with relatively little attention paid to other mathematical topics (Bae et al., 2015; Berciano Alcaraz et al., 2022; Gevarter et al., 2016; King et al., 2016). Specifically, there is a lack of understanding about how students with ASD, both with and without intellectual disabilities, learn algebra. Given the growing emphasis on algebraic concepts in primary education curricula across various countries (Ministerio de Educación y Formación Profesional de España [MEFP], 2022, National Council of Teachers of Mathematics NCTM, 2000), and the increasing number of students with ASD in general education (McDonald et al., 2019), it is crucial to better our understanding of how these students develop their algebraic skills. Therefore, this study aims to examine how students with ASD in primary education approach tasks that involve algebraic concepts.

Early algebra through generalization tasks

Generalization is a crucial cognitive process in mathematics, particularly for students in primary education (Pinto et al., 2022). Generalizing in mathematics involves identifying and abstracting characteristics and regularities from specific cases, in order to arrive at rules, laws, theorems, and other mathematical concepts. Algebra, in particular, requires a strong ability to generalize. According to Kieran (2004), algebraic thinking in the early grades involves developing ways of thinking such as analyzing relationships between quantities, noticing structure, studying change, generalizing, problem-solving, modeling, justifying, proving, and predicting (p. 149).

Traditionally, algebra has been introduced in secondary education. However, recent research on learning suggests that there are many reasons to develop algebraic ideas earlier in a student's education (Kieran, 2018). As a result, more and more countries are including algebraic content in primary education curricula (MEFP, 2022, National Council of Teachers of Mathematics [NCTM], 2000). This approach is known as 'early algebra' and aims to introduce algebraic thinking in a meaningful and accessible way for primary school students (Carraher et al., 2008). Early algebra involves activities that do not require the use of letters or algebraic symbols (Kieran, 2004). Several investigations into early algebra have analyzed the role of generalization from various perspectives. Specifically, researchers have observed students' ability to abstract structures from arithmetic expressions and to express algebraic rules for numerical, geometric patterns, and functional relationships (Cañadas et al., 2016). To achieve generalization there are certain mathematical skills to be developed, namely organizing information, identifying structures, establishing a conjecture and validating it. In the case of visual linear pattern tasks, the focus of this paper, the students must learn to compare the different figures holistically, and then decompose them into parts, so as to observe what changes at each step and what remains the same. This implies visualization and flexibility of thought. They must also reflect numerically what they observe in the figure in order to construct an additive or multiplicative relationship, the regularity of which can be checked in other steps of the sequence (Moss & London McNab, 2011).

The NCTM standards highlight the importance of effective math reasoning, which involves the ability to notice patterns and structures in both real-world situations and symbolic objects. This reasoning enables the formation of generalizations by abstracting ideas and relationships. As such, the study of patterns, relationships, and functions is proposed as a cross-cutting content from early childhood education to the final years of school (National Council of Teachers of Mathematics NCTM, 2000). In Spain, where this research is conducted, the mathematics curriculum for primary education also proposes that students in all grades should be able to describe, analyze, and find patterns in numerical, geometric, and functional contexts by the end of this level (MEFP, 2022). In this paper, we focus on generalization from a functional approach to early algebra, where functions are the main mathematical content. Although functions are not explicitly worked on in primary education, we propose tasks where functions represent the regularity between two variables (from now on,



1. Draw the figure at positions 4. How many points did you need to draw the figure?
2. Draw the figure at positions 5. How many points did you need to draw each figure?
3. How many points do we need to draw the figure at position 10?
4. How many points do we need to draw the figure at position 25?
5. How many points do we need to draw the figure at position 12?
6. How many points do we need to draw the figure at any position?
7. What position in the sequence corresponds to the figure consisting of 11 points?

Figure 1. Example of a generalization task involving the function y = n + 2.

generalization tasks). Specifically, we focus on linear functions (i.e., y = a·n + b, where y, n, a, and b are whole numbers).

As an example, Figure 1 displays a generalization task involving the function y = n + 2. The figure on the left illustrates the first terms of a growing sequence of whole numbers, represented by figures, which in this case are configurations of points. In this context, the independent variable (*n*) refers to the position of the drawing within the sequence, while the dependent variable (*y*) indicates the number of points in that position.

The information presented can be organized into a table that relates the data for both variables (see Table 1).

For this type of task, various task variables are taken into account in the design of the proposed work for the students. Typically, we begin with particular cases of different types that are adapted to the children involved. We consider the numbers with which they are familiar, so that the calculations do not pose an additional difficulty. Additionally, we start with consecutive particular cases and then progress to nonconsecutive ones. Passing through distant particular cases, we then move on to asking for the general rule. When the roles of the variables are exchanged, and thus the value of the dependent variable is known and the value of the independent variable is unknown, we speak of the inverse form of the function (Merino et al., 2013). In the given example, it would involve finding the position that corresponds to a given number of points. This is observed in question 7 of the task presented in Figure 1, which asks for the position that corresponds to the figure if 11 points are used to draw it.

The main objective of these tasks is to recognize and identify the regularity presented in the given sequence. This recognition can be achieved through different actions: (a) identifying regularity in consecutive input values (questions 1 and 2 in Figure 1), (b) recognizing regularity in any term of the sequence (nonconsecutive input values; questions 3, 4, 5, and 7 in Figure 1), and (c) using the regularity to determine a rule to find the number of points in any term of the sequence (generalization; question 6 in Figure 1).

Functional relationships between variables can be identified through recursive, correspondence, or covariation patterns (Smith, 2003). Recursive involves observing variation within a sequence of values and describes the change in values within the same variable (in this example, the value of y increases by

Table 1. Table to represent the data for y = n + 2.

Position	Number of points
1	3
2	4
3	5
...	...

one). In particular, one asks oneself: 'What do I need to do to the current stage to reach the next one?.' The relationships between two patterns through a rule that helps find 'y' based on a specific 'x' value (like 'y = 4× + 1'), is called a "correspondence description" of the relationship. "Covariation" refers to how quantities in two sets of data change simultaneously, highlighting the relationship between changes in one pattern and changes in another pattern (Confrey & Smith, 1995; Cooney et al., 2010; Panorkou & Maloney, 2016). These functional relationships can be identified in students' responses when working with questions involving specific values and/or when generalizing the relationship (e.g., Pinto & Cañadas, 2021; Pinto et al., 2022). Typically developing elementary students have shown their ability to think about the relationship between two sets of data and to express this relationship in a very abstract way (e.g., Pinto & Cañadas, 2021; Warren & Cooper, 2008).

These functional relationships are typically expressed through different types of representations. Representations are the means by which students organize and express the relationships they identify, and understand, analyze, explain, predict or justify the way variables are related (Pinto & Cañadas, 2021). We also emphasize external representations, which are perceptible through the senses and provide information on how students carry out their work (Merino et al., 2013). There are various types of representations through which students can express generalization, including natural language, pictorial (such as simple or detailed drawings), numerical (using numerical symbolism), algebraic notation, tabular (using tables), graphic (employing graphs), manipulative (using materials such as blocks, tokens or fingers) and gestures (Carraher et al., 2008; Merino et al., 2013; Pinto & Cañadas, 2021).

In the presented task example, we observe natural language, numerical, pictorial (dot configurations), and tabular representations. While algebraic symbolism is the primary representation used in algebra to express unknown quantities and generality, it requires a level of abstraction that not all elementary school students have reached.

Previous research in early algebra

There are numerous studies that show evidence of generalization in tasks that aim to promote algebraic thinking and involve functions in early elementary school (e.g., Blanton et al., 2017; Castro et al., 2017; Pinto & Cañadas, 2021). For example, Castro et al. (2017) showed the ability to identify functional relationships of *correspondence* and *covariation* in 5- and 6-year-old children, with a greater emphasis on developing intuitive ideas when the functional relationship was the identity relationship. Pinto and Cañadas (2021) found evidence of functional relationships in 10- and 11-year-old students when working with tasks involving the function $f(x) = 2\times+6$ and generalized that relationship.

Recent studies on early algebra have emphasized the importance of representations as one of the central practices to promote algebraic thinking (e.g., Narváez & Cañadas, 2023; Pinto & Cañadas, 2021; Pinto et al., 2022; Torres et al., 2022). In the early grades, some students evidenced generalization through verbal or even numerical expressions. Torres et al. (2022) highlighted that second graders' organization of data in tables facilitated the identification of regularities between variables involved in the function. In contrast, Pinto et al. (2022) found that not all 8–9-year-old students generalize the relations involved in tasks involving functional relations, and most of those who do generalize express it verbally. Regarding the supports that have been shown to be effective in generalizing among elementary school students, Narváez and Cañadas (2023) described the generalization evidenced by 6-, 7- and 8-year-old students when solving a task involving a linear function. After conducting mediations with students who experienced difficulties, they were able to generalize. Mediation processes (Ponte et al., 2013) have been successfully used in previous research to facilitate generalization (Mata-Pereira & Ponte, 2017). The mediator used actions such as: (a) inviting, aimed at initiating a discussion; (b) supporting/guiding, aimed at guiding students to solve a task through questions or observations and that point, explicitly or implicitly, the way for them to follow; (c) informing/suggesting, when the teacher introduces information, gives suggestions, presents

arguments or validates students' answers; and (d) challenging, when the teacher seeks for students to take on the role of producing new representations, interpreting a statement, making connections or formulating a reasoning or evaluation.

Algebraic thinking in students with ASD

Recent reviews have highlighted the lack of research on algebraic content for students with special needs (Stephens et al., 2022), particularly for those with ASD (Spooner et al., 2019). The introduction of algebra can be challenging for students with ASD, as it requires understanding abstract concepts and procedures, which can be difficult for individuals with deficits in abstract reasoning and can lead to inflexibility of thought (Cleary & Hart Barnett, 2015; Hart Barnett & Cleary, 2019). To date, there has been little research on early algebra for students with ASD (Hart Barnett & Cleary, 2019). For example, Hart Barnett and Cleary (2019) showed the effectiveness of an instruction based on visual supports for teaching algebraic equations to a student with high functioning ASD. Regarding early algebraic thinking, Goñi-Cervera et al. (2022) study identified and described strategies and representations used by 26 elementary school students with ASD when performing a task involving geometric patterns. The strategies included: (a) simply responding; (b) modeling with manipulatives; (c) drawing; (d) counting; and (e) operating, with the latter being the most frequent, represented verbally or symbolically, followed by drawing. Few students were able to reach higher levels of generalization, indicating the need to address this issue in classroom tasks. Given the lack of research in this area, we aimed to evaluate an instructional sequence that could support the development of generalization in early algebra tasks for elementary school students with ASD.

Cleary and Hart Barnett (2015) suggested that it is crucial to develop evidence-based classroom implementation strategies to assist teachers in addressing the mathematical deficits of students with ASD, while considering their strengths and weaknesses. Notable strengths include their frequent ability to develop mathematical learning through the use of visual representations (Grandin, 1995; Iuculano et al., 2014; Rockwell et al., 2011; Root et al., 2017). Specifically, C. Wong et al. (2015) indicated that visual representations provide cues to guide students with ASD when performing academic tasks.

The goal of this study is to evaluate the effectiveness of an instruction based on multiple representations in promoting early algebraic thinking in students with ASD. Specifically, we aim to answer the following research questions: (1) Can the use of multiple representations with mediation improve the performance of students with ASD on generalization tasks in functional contexts? (2) Will the improvements in task performance be transferred to everyday situations involving regularities, and will they be maintained over time? and (3) How do students with ASD perform on tasks that require distinguishing between questions involving consecutive terms, nonconsecutive terms, and the general rule?

Method

Participants and setting

The participants were three students selected from a broader project related to the mathematical learning of children with autism, whose aim was to describe the mathematical skills of students with ASD and their relationship with cognitive variables such as executive functions (Polo-Blanco et al., 2024). In this project, the participants had been recruited through associations of people with disabilities or autism, regular and special schools, guidance teams, and hospital outpatient clinics. For the present study, the following inclusion criteria were considered: being diagnosed with ASD according to the Diagnostic and Statistical Manual of Mental Disorders, 5th edition (DSM-5, American Psychiatric Association [APA], 2013), having a math equivalence age of at least 6 years on the Test of Early Mathematics Ability TEMA-3 (Ginsburg et al., 2007) to ensure prerequisite

Table 2. Participants data.

Student	Age	IQ[1]	Math Age[2]	Grade	Math Curricula[3]	Specialist Support[4]
A	9,41	88	6,83	4th	2nd-3rd	6
B	7,25	87	6,66	2nd	2nd	4
C	7,41	96	5,5	2nd	2nd	2

Note. [1]Measured by WISC-V (Wechsler, 2015); [2]Measured by TEMA-3 (Ginsburg et al., 2007); [3]Mathematic curricula followed by the student at school; [4]Hours per week at school.

knowledge of addition and subtraction number facts, and satisfactory performance on the pre-screening measure. The pre-screening measure assessed the participants' ability to (1) complete repetitive pattern tasks with one or two attributes (e. g., continue the pattern: red circle, blue circle, red circle, blue circle . . .), and solve generalization tasks involving the $y = n + 1$, $y = n + 2$, $y = n + 3$, and $y = 2{\cdot}n$ functions. The goal of the first part of the pre-screening measure tests was to ensure that the participants were capable of working with repetitive patterns independently. The goal of the second part was to ensure that the participants had not acquired the ability to work with generalization tasks involving functions. A participant achieved satisfactory performance on the pre-screening measure if they completed the first part with more than 90% accuracy and the second part with no more than 25% accuracy.

The three participants were Spanish males diagnosed with ASD and knew the instructor from a previous study conducted on additive comparison problem solving (Goñi-Cervera et al., 2024). The instructor, who is the first author of the paper, had a strong connection with the school, its teachers, and the families of the participants due to her prior involvement in the mentioned study. She possessed a bachelor's degree in primary education and a master's degree in mathematics education. During the study, she was a second-year Ph.D. student working on her thesis related to mathematics education for students with autism.

The participants had not been exposed to generalization in visual growing patterns in school. All three participants attended the same general education school located in a medium-high socioeconomic context in the north of Spain. In addition, Student B and C attended the same classroom. Table 2 shows some characteristics of the participants.

Dependent and independent variables

The dependent variable was the number of generalization tasks correctly solved by the students autonomously in each evaluation session. The independent variable was multiple representation with mediation intervention detailed in the procedure section.

Design and data collection

A single-case, multiple-probe across students' design (Horner & Baer, 1978) was followed to assess the effectiveness of the intervention. The phases of the experiment were: baseline, instruction on generalization tasks, follow-up, transfer to real-life contexts and maintenance.

Probes and scoring

During all phases of the evaluation, a probe was used which contained a graphical representation of the first three steps of a pattern and seven questions as previously described, in line with NCTM Standards (Phillips et al., 1991). In the case of the training phase, this probe was conducted after the intervention session. The questions of the probe involved consecutive and nonconsecutive terms, inverse relationships, and the general term (refer to Figure 1 for more details). Questions 1, 2, 3, 4, 5, and 7, which pertained to particular terms, could each score a maximum of 1 point. Question 6, which involved the general term, could earn a maximum of 2 points. If a student correctly expressed the particular

relationship by providing an example, they would be awarded 1 point (e.g., 1000 + 1). If they also expressed the general rule, 2 points would be given. The maximum score per probe was therefore 8 points.

As in previous works on algebraic thinking and ASD (Goñi-Cervera et al., 2022), the tasks were deemed suitable for the participants because the representations could effectively cater to the strong visual processing often observed in these students. The questions about different terms of the patterns were adapted from previous works on algebraic thinking with typically developing students of similar ages (Pinto & Cañadas, 2021). They were considered well-suited for the study's participants, in accordance with the Spanish primary curriculum's emphasis on students, from the first grade onward, working to 'recognize and describe regularities and patterns in numerical, geometric, and functional contexts' (MEFP, 2022).

Procedure

In order to demonstrate the relationship between instruction and student performance in solving generalization tasks, the instruction was introduced to students sequentially over time. The first student began the instruction after demonstrating a stable baseline (at least two consecutive sessions with the same score). Once a noticeable improvement in performance was observed over at least two sessions, the instruction was introduced to the second student, provided that they had also previously shown a stable baseline. This process was repeated for the third student.

The various phases of the study are outlined below.

Baseline

During each baseline session, every student was presented with a probe similar to the one depicted in Figure 1, featuring generalization tasks involving functions (such as $y = n + 1$, $y = n + 2$). The instructor would introduce the task by saying: 'Now we are going to build figures. Observe how this figure grows. Let's see what we are required to do.' For each baseline task, the students were provided with a worksheet and a pencil, and given approximately 15 minutes to complete it. The instructor did not interfere except to assist with reading the questions or to encourage participation if the students were unfocused or unresponsive.

Training

A study was conducted using an intervention approach that involved multiple representations and a guideline sheet with mediation for three students. Multiple representations were employed because previous research has shown them to be effective in helping elementary school students solve generalization tasks (Narváez & Cañadas, 2023; Pinto & Cañadas, 2021; Pinto et al., 2022; Torres et al., 2022; Warren & Cooper, 2008). The representations used in this study included manipulatives (multilink, sticks, and dot stickers), pictorial representations (visual patterns), and tabular representations. In addition, a guideline sheet was used (see Figure 2, left) as a tool to implement scaffolded learning (Spooner et al., 2019). The purpose of the guideline sheet was to shift control of discriminative stimuli from the instructor and thereby promote students' autonomy, which is a key focus when working with students with disabilities (Spooner et al., 2017). In the case of students with ASD, a guideline sheet with visual supports also builds on their strengths in relation to routine structures and visual processing, while compensating for potential weaknesses in communication and executive functions, such as planning or working memory (Polo-Blanco et al., 2024; Root et al., 2017; Spooner et al., 2017).

Each training session was designed to last a maximum of 45 minutes and was divided into two parts: (1) intervention phase and (2) evaluation phase. Figure 2 shows the materials available during training sessions. At the beginning of each session, the instructor provided the students with the guideline sheet, which they read together. Following the steps provided, the instructor invited and guided the students in solving the worksheet.

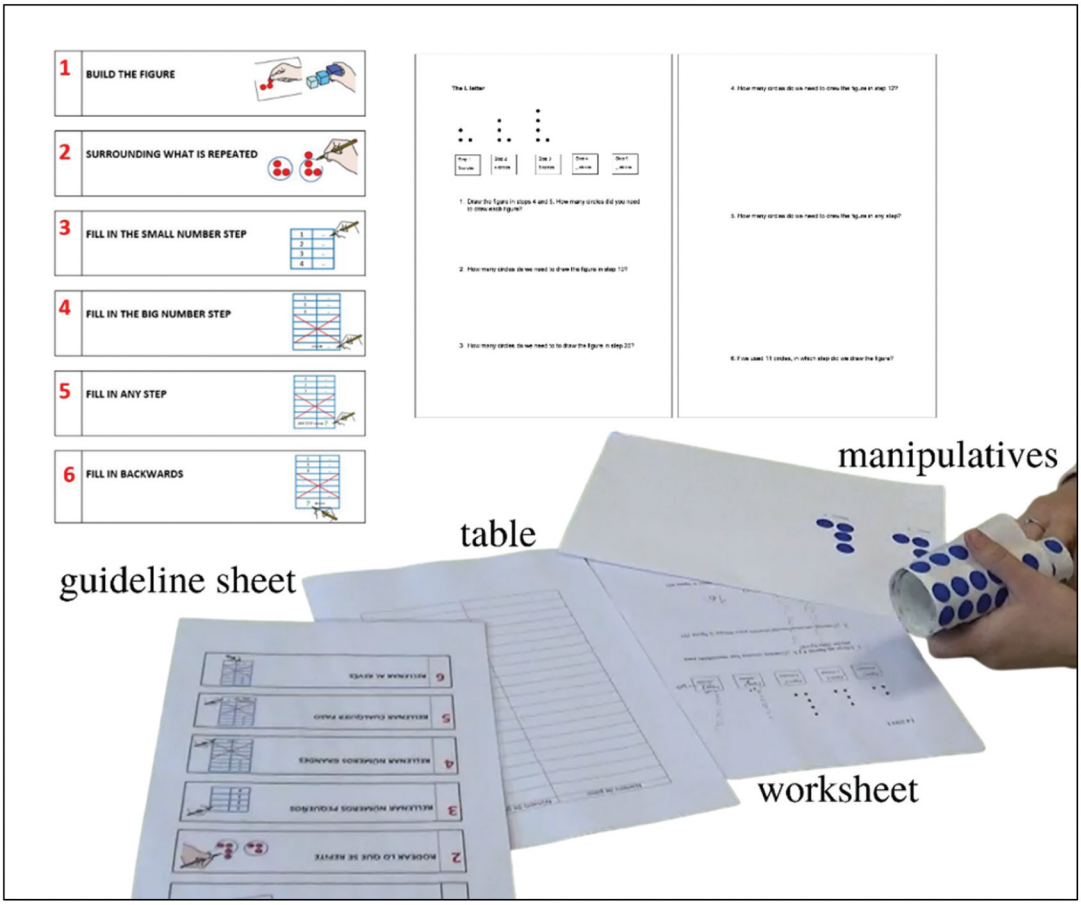


Figure 2. Materials available during training phase.

The intervention phase was carried out using a mediation process (Ponte et al., 2013). As mentioned above, the mediation process involved actions such as inviting, informing/suggesting, supporting/guiding and challenging the students depending on their attitudes toward the task.

For example, when working on the task in Figure 1, the instructor first invited the students to use the materials and suggested constructing the different steps in the figure to answer questions 1 and 2, which involved consecutive cases (therefore using the pictorial and/or manipulative representations). The primary guiding action was to propose the use of a 'guideline sheet,' which included several steps: (1) building the figure, (2) circling what is repeated, (3) filling in the small number step, (4) filling in the big number step, (5) filling in any step, and (6) filling in backwards. These steps were designed to develop the actions involved in the generalization processes (Dreyfus, 1991) and to assist with the resolution by distinguishing between different types of terms (consecutive, nonconsecutive, and general terms). Steps (1) and (2) aimed to help the students identify the pattern's regularity, step (3) to apply that relationship to consecutive terms by relying on the table, step (4) to help transition from recursive strategies to establishing relationships between variables, step (5) to find a rule for the general term, and step (6) to apply that rule in reverse. The guideline sheet was written in clear handwriting and accompanied by pictograms to aid comprehension. The instructor questioned the students at certain times to help them reflect on the task if she observed errors or difficulties. Finally, for questions involving the general or inverse term, the instructor challenged the students to formulate

Table 3. Examples of the use of different representations and mediation actions when solving the generalization task in figure during the intervention phase.

Task question	Representation	Examples of guiding questions during mediation supported by the guideline sheet
1, 2	Pictorial	Invite: Let's see what they're asking us to do. Inform/suggest: We can try using these dot stickers. Support/guide: Look at the guideline sheet, it tells us to circle what is repeated.
3	Tabular and pictorial	Invite: Let's refer to the guideline sheet. Inform/suggest: We can write on the chart the steps we already know. Would you prefer to build the figure using dot stickers? Support/guide: For example, in step 4 we needed 5 dot stickers... let's add it to the board.
4, 5	Tabular	Challenge: Let's skip all these steps on the board. Let's cross them out. Suggest: Notice how the figure was growing before. Challenge: Will we need more rubber bands or fewer rubber bands? How many more do we need? Suggest: It looks like we need more rubber bands.
6	Tabular	Support/guide: We're being asked for any steps. Challenge: How many dot stickers would we need in step 100? What about in a made-up step? Can you think of another number? If you were in that step, how many circles does the figure have? Suggest: Let's try with a specific number: do you think it will be two more? three more? what do you think?
7	Tabular and pictorial	Support/guide: Be careful now, take a closer look at the table. Inform/suggest: Now the question is the other way around. They give us the number of dot stickers and ask us for the step of the figure. [If the student answered incorrectly] Do you want to build the figure using the 7 dot stickers?

reasoning that would enable them to generalize the relationship to other terms in the sequence (including, primarily, natural language representation, but also numerical or even algebraic notation).

Table 3 presents the guiding questions associated with the mediation process for solving the task displayed in Figure 1 during the intervention phase. The type of representation used for the resolution of each task question is also specified.

In the evaluation phase, students completed a task similar to the baseline or intervention phase tasks, with the materials as described in Figure 2 available, but without any mediation from the instructor. Training included a minimum of four sessions per student and lasted until at least 75% accuracy was achieved in two consecutive evaluation sessions.

Follow-up

Two to three follow-up sessions were conducted, each lasting approximately 15 minutes. In order to compare performance across different phases under similar conditions and thus demonstrate students' improvement (Horner & Baer, 1978), during sessions in this phase students completed independently a similar probe than in the baseline sessions, that is, without having at his disposal the guideline sheet or the representations used during the instruction. From an educational standpoint, it is crucial to gradually decrease support for students with learning disabilities when they are prepared to advance, thereby seeking to align the instruction with the standard math practices employed in general education (Spooner et al., 2017).

Transfer to real-life context tasks

During the transfer phase, students performed two generalization tasks with a similar structure to those performed in the previous phases, but contextualized in real-life situations and presented in verbal format (see Table 4 and Figure 3).

In the first generalization transfer task, called 'theme park,' children were tasked to calculate the amount of money required to ride a given number of attractions at a theme park. They were given the information that the park entrance costs 2 euros and each attraction costs 1 euro (refer to Figure 3, left). The second generalization task, called the 'game of chairs,' required them to determine the number of children who could participate in the game based on the number of chairs available. The

Table 4. Transfer to real-life context tasks presented.

Task	Function	y	n	constant	Visual aid
Theme park	y = n + 2	Money needed	Number of attractions	Entrance fee (2€)	Figure 3, left
Game of chairs	y = n + 1	Number of children that can play	Number of chairs	A child is left without a chair	Figure 3, right

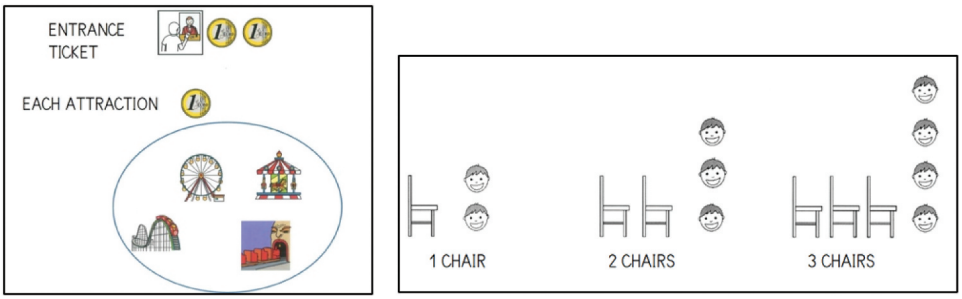


Figure 3. Visual supports for transfer to real-life context tasks.

task involved a drawing of one chair with two children, two chairs with three children, and three chairs with four children (refer to Figure 3, right). These tasks were adapted from activities designed for children aged 6 to 11 years old, as presented in various studies conducted under the research project 'Algebraic thinking in early childhood education and primary education,' of the University of Granada (https://pensamientoalgebraico.es/en/).

Maintenance

A maintenance session was held with each student five to six weeks after completion of the instruction. The students solved independently a probe similar to those in the baseline and follow-up phases.

Internal validity

All the sessions were videotaped, and inter-rater and procedural agreement data were collected randomly from all phases. A university teacher specialized in mathematics education who did not know the hypotheses of the study, reevaluated about 39% of the students' performance in the evaluation phases. Specifically, about 29% of baseline's sessions, 50% of training's, 25% of follow-up's, 50% of transfer's and 67% of maintenance's sessions were reevaluated. Inter-rater agreement was calculated for each phase dividing the number of agreements by the total number of data and the result was multiplied by 100.

Procedural reliability measured the instructor's performance with respect to the planned behaviors, which were: (1) provides the agreed number of problems in each session; (2) provides the necessary supports such as multi-presentation material in the training phase; (3) provides a phase of mediation (inviting, supporting/guiding, informing/suggesting, and challenging) when necessary; (4) allows students to solve the tasks autonomously; and (5) congratulates students at the end of each session. Procedural agreement was calculated by dividing the number of observed teacher behaviors by the number of planned behaviors and multiplying by 100 for approximately 39% of the sessions.

Social validity

Social validity is an essential aspect that deals with the social significance and social appropriateness of the effects of interventions (Gresham, 1983). For this reason, examining social validity to meet the concerns of students, family members or teachers is important as part of applied research.

At the outset of the study, the students were surveyed on their interest and motivation in the field of mathematics, while their families and teachers were surveyed on the students' attitudes toward mathematics and their performance in the subject. Upon completion of the instruction, the instructor provided the families and school teachers with a comprehensive report on each student's progress during the intervention, as well as samples of their task resolutions at different intervals. Follow-up interviews with the students focused on their level of satisfaction with the instruction and whether they wished to continue working with the instructor. Additionally, families and teachers were surveyed on their perceptions of the intervention and the materials used, with the latter being asked to identify the most effective instructional materials and whether this experience could be extended to the classroom to promote the inclusion of ASD students.

Results

Figure 4 shows the number of points obtained by the three students in solving generalization tasks during the evaluation phase of each session. The maximum score was 8 points, which was achieved by answering all 7 questions correctly, and earning two points for question 6 by providing the general rule correctly. In total, 15 sessions were held with students A and C and 16 sessions with Student B. The results for each student in each condition are described below.

We employed guidelines for visual analysis of SCRD data (Kratochwill et al., 2013), to examine the data for various aspects, including level, trend, variability, immediacy of effect, overlap, and consistency of data patterns. The results showed an immediate improvement for all three students, with a mean accuracy of 79% during the training phase, 84% in the follow-up phase, and 83% during the maintenance phase. During the baseline condition, all three participants displayed a stable pattern of responding. However, during the intervention phase, all three participants demonstrated a change in level and an increasing trend, with no overlap with their baseline performance. Visual analysis of the graph revealed a functional relationship between the intervention and the number of correct answers obtained in generalization tasks. In Table 5, we present the results of the different steps of the task (Figure 1) in terms of the mean percentages of correct answers, categorized by type of term: consecutive (questions 1, 2), nonconsecutive (questions 3, 4, 5 and 7), and general (question 6) for each student in each condition.

Student A

Student A had a mean baseline performance of 15% (see Figure 4 and Table 5). During this phase, Student A correctly answered 50% of the questions involving consecutive terms arriving at the solution using drawing strategies. In his drawing-based strategies, both correct and incorrect, he always imitated the shape of the pattern and identified which part of the pattern (e.g. of the two segments of an 'L,' see Figure 1) increased from one step to the next. The errors were due to drawing fewer dots, especially for the function $y = n + 1$. For the nonconsecutive cases, he answered incorrectly on almost all occasions, providing the same number as the step as the answer. He did not correctly answer any question involving the general term, always answering 'the previous term.' He showed an immediate increase in the resolution of generalization tasks upon introduction of the intervention, with a mean intervention performance of 84% during training phase. There was an accelerating trend across the instructional sessions from the second session on. If we distinguish by type of term, he also showed an immediate increase upon introduction of the intervention, with a mean intervention performance of 100% for consecutive terms, 88% for nonconsecutive terms and 63% for the general term.

During the intervention phase, the instructor built on the fact that the student consistent identified the pattern in consecutive cases during the baseline phase, and encouraged him to first represent these terms using drawings of manipulatives, and then transfer what he had modeled to the table. Furthermore, the instructor motivated the student to make conjectures about what to enter in the table when extending the terms to nonconsecutive cases, aiming to enhance his ability to identify

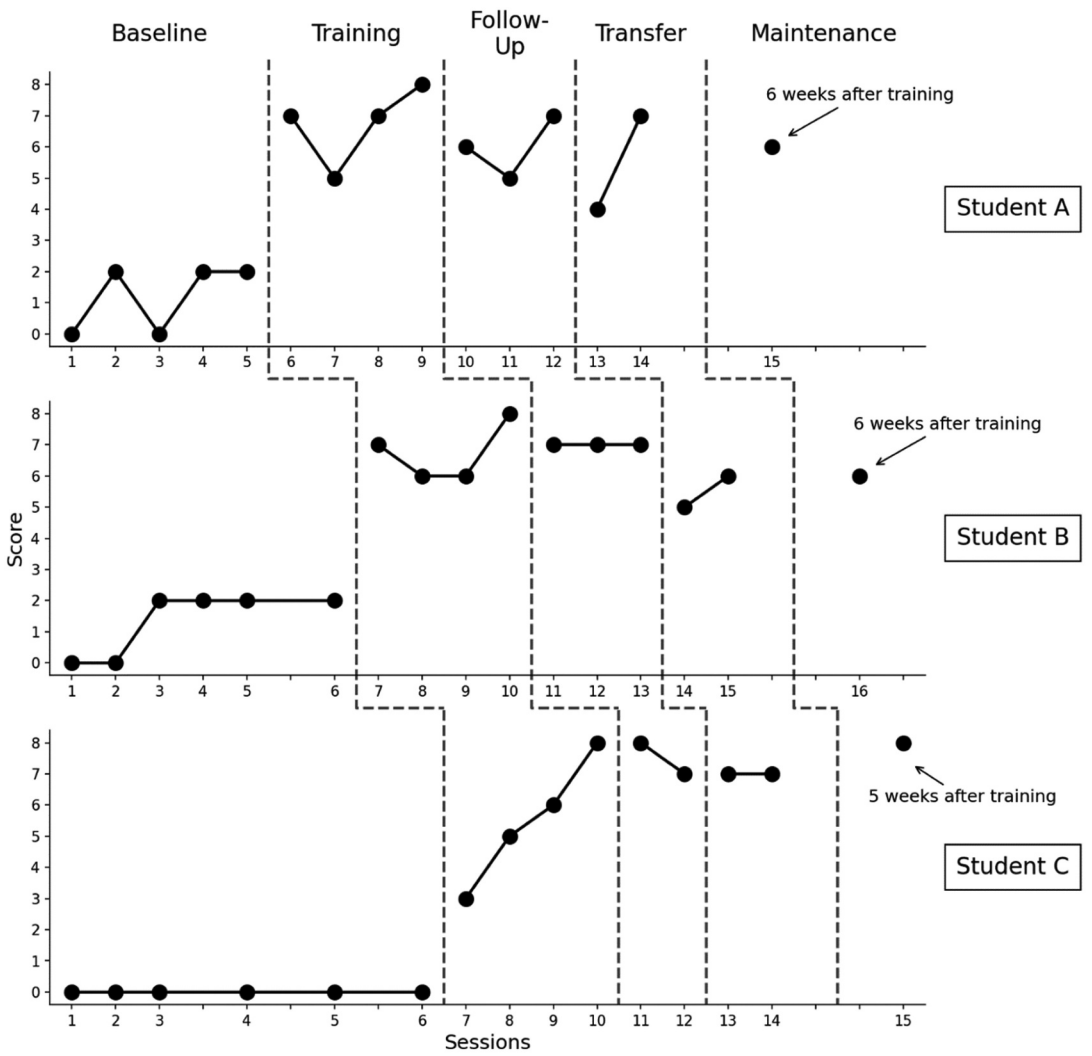


Figure 4. Number of points obtained by the three students in solving generalization tasks.

Table 5. Percentage of successes by session, type of terms and student.

	Consecutive Terms				Non-Consecutive Terms				General Term			
	B	T	F	M	B	T	F	M	B	T	F	M
Student A	50	100	100	100	5	88	92	100	0	63	17	0
Student B	67	100	83	100	0	100	92	100	0	38	83	0
Student C	0	75	75	100	0	81	100	100	0	38	100	100

Note. B=Baseline; T=Training; F=Follow-Up; M=Maintenance.

underlying relationships. For example, the first training session, in which the function involved was y = n + 1, he first modeled the pattern for the first nonconsecutive terms, and then wrote down the first 10 terms in the table. When asked for the number of points needed for item 25, he answered spontaneously: '26.' To facilitate the understanding of the question: 'How many stickers do we need to build the figure in any position?' the instructor, during the intervention phase, asked him for step 400 and student A answered: 'That would be 401.' In the second training session, where the function

involved was y = n + 2, student A began associating the number of the position of the figure with the points needed to build it consistently utilizing the table and relying on a recursive relation. For example, when asked: 'How many points do we need to build the figure in position 25?' Student A completed the table up to position 25, and then answered: '27. I have known it by counting by twos,' evidencing a functional correspondence relationship supported by the tabular representation. Next, he erroneously solved the question involving the inverse relationship, performing it as if it were a direct case question. When asked about nonconsecutive terms, Student A employed a recursive strategy based on the previous terms and relying exclusively on the table. For example, for y = n + 3, when asked about the points needed to construct the figure 'in any position,' he looked at the table, where he had noted that for position 25 he would need 28 points, and said, 'Counting one at a time. For example, for position 26, I would need 29 points, because that's one more (pointing out on the chart the 28 points he had noted for position 25).' In this particular example, Student A again shows a functional recursive relationship by relying on the table.

To assist Student A in articulating the general rule and fostering an understanding of the relationship between the variables, the instructor often employed a mediation approach during the non-evaluable training sessions (intervention phase). To achieve this, she utilized the table and systematically examined the consecutive terms mentioned earlier, guiding Student A to express the relationship between the variables. She encouraged Student A to closely scrutinize the interplay between the columns in the table, prompting him to consider whether the number on the right was 'one more,' 'two more,' or 'three more' than the number on the left.

At the follow up phase we observed a slight variability, with a mean performance of 75% and no overlap of data between baseline and follow-up. This variability is due to the fact that sometimes he answered incorrectly the questions involving the general term although he answered correctly 100% of the questions involving consecutive terms, and a performance of 92% in the questions involving nonconsecutive terms (see Table 5). During these follow-up sessions, in which neither the table nor the guideline sheet was provided, Student A drew a table in the space provided in the worksheet and showed similar reasoning to the training sessions, again evidencing a functional recursive relationship.

During transfer to real-life context, student A showed difficulties (50% success rate) in understanding the first questions of the first task ('theme park'), but then he was able to express the nonconsecutive cases of the following questions and to particularize to other invented cases, without generalizing to any number or showing understanding in the inverse relation question. In the second task ('game of chairs'), he exhibited a correct response rate of 87%. To respond, he created a table that he completed and correctly answered the given cases, relying on it. In addition, he identified a functional correspondence relationship, responding correctly to the general case by saying that for any number of chairs he will be able to play 'one more child than chairs.'

He maintained what he learned five weeks after the last training session, exhibiting a correct response rate of 100% for consecutive and nonconsecutive terms, but without correctly answering the question involving the general term. For this last question, he answered: 'the necessary points of a table,' alluding to the need to work with a tabular representation to answer the questions.

In summary, during the instructional phase, the table proved to be the most effective representation for transitioning to nonconsecutive terms. Although he exhibited some resistance to abandoning it, relying on the table facilitated his recognition of the recursive relationship between variables and facilitated his progress toward generalization.

Student B

The student B had a mean baseline performance of 17%. During this phase, the correct answers corresponded to some questions about consecutive terms which he approached with drawing strategies. He identified the pattern for consecutive terms for all shapes except for letters formed by dots (see for example Figure 1), where he interpreted both segments of each letter as growing. Curiously, in the questions about nonconsecutive terms, he identified correctly the number of

parts to add in most patterns, but subtracted them from the step instead of adding them (i.e. if in each step one point was added, for step 10 he answered 9, and if two were added, he answered 8). For the general case, he provided apparently arbitrary numbers without providing explanation of his thinking. As illustrated in Figure 4, he showed an immediate increase in the resolution of generalization tasks upon introduction of the intervention, with a mean intervention performance of 84%. There was an accelerating trend across the instructional sessions from the second session on, without overlapping with the baseline. If we distinguish by type of term, he had a mean intervention performance of 100% for consecutive and nonconsecutive terms, and 38% for the general term (see Table 5).

During the intervention phase of training, student B accepted the use of the materials for the representation of the figures. The instructor initially utilized the patterns in which the student had correctly responded to consecutive terms in the baseline phase. Using manipulatives, primarily sticks and dot stickers, she guided him to represent the terms and identify the recursive relationship. The student quickly grasped the relationship, applying it correctly to other terms by adding it to the previous step, in contrast to his baseline approach of subtraction. For example, for the function y = n + 1, for step 10 he correctly answered 11 (instead of 9, as he had done in baseline) arguing: 'because I'm skipping one.' He also applied the rule correctly for the inverse term, by claiming that 'if we subtract 1 from 7, it goes down to level 6.'

In the last training session, involving the function y = n + n, the instructor encouraged Student B to rely on the tabular representation, writing down the particular cases. For example, to answer the nonconsecutive term $n = 10$, the student wrote down on the right side of the column the number 20. When the instructor asked him for his reasoning, Student B answered, looking at the table and using natural language, that it was '10 + 10.' To help him to find the general rule, the instructor again leaned on the table, guiding him to recognize that from one column to the next, the same number was added to itself. After going through several particular cases, the instructor asked Student B how many sticks it would take to draw the figure in any step and challenged him to put in the left column of the table something, like a star, that represented 'any step.' But then Student B, spontaneously said that if it was an X instead of a star, the answer would be X + X. The student proceeded to provide examples in natural language, saying, "If I put 'bicycle,' it would be 'bicycle plus bicycle," and he filled in the table by drawing the bicycles. He continued the task by inventing words, playfully asking, 'And for the yoyo step?' before responding to himself with laughter, 'yoyo + yoyo.' This demonstrated his grasp of the general rule and his ability to generalize through a functional correspondence relation. Moreover, for the inverse term (12 points) he answered that it would be position 6 because 12 is '6 + 6,' showing again a functional correspondence relation.

During the follow-up phase, Student B's performance showed a slight improvement, achieving 88%, without overlapping with the baseline (Figure 4). In the questions involving consecutive terms, as no materials were available and the student had to resort to drawing, the errors were similar to those shown in the baseline. An increase in performance was observed in the questions involving the general term, going from a mean performance of 38% in training to 83% in the follow up sessions. In these questions, student B generalized through a combination of pictorial and numerical representation, using drawings of invented objects, again showing functional correspondence relationships. For example, for the task involving the function y = n + 2 and following the generalizations of the previous training session, he responded to the general term by drawing a wing and adding '+2.'

In the transfer tasks, he responded correctly to most of the particular cases, including the inverse, but not to the general term. When answering the inverse term, he showed evidence of generalizing by means of a functional correspondence relationship, relating the two variables, although omitting them when expressing the generalization by means of natural language. For example, in the second task ('game of chairs'), he said 'because [children] are moved from one [chair].'

He maintained what he learned 6 weeks after the last training session, exhibiting a correct response rate of 100% for consecutive and nonconsecutive terms, but without correctly answering the question involving the general term.

Student C

Student C had a mean baseline performance of 0%. During this phase, he applied drawing strategies to all terms (consecutive, nonconsecutive, inverse and general), although they did not lead him to the correct answer in any case. His patterns mimicked the shape of the provided pattern and increased the number of parts (dots, etc.) in each step, but the quantities he added did not maintain any regularity. Furthermore, in the case of dot letters (see, for example, the letter L, Figure 1), he increased the number of dots in all segments without identifying the constant part. He showed an immediate increase in solving generalization tasks upon introduction of the intervention, with a mean intervention performance of 69% (Figure 4). There was an accelerating trend across the instructional sessions without overlapping with baseline. If we distinguish by type of term, he showed an immediate increase, especially for consecutive and nonconsecutive terms, with a mean intervention performance of 75% and 81%, respectively (Table 5).

During the instruction, student C accepted the use of materials and relied on the guideline sheet to solve the tasks. During the initial sessions, the student lacked motivation, so the instructor's mediation focused on providing encouragement and guidance to help the student become more engaged. In particular, during the instructional phase, the instructor specifically targeted the fact that, in the baseline phase, the student hadn't identified the regularities in the patterns. To address this, the instructor concentrated on demonstrating how she personally constructed the patterns for the initial terms. She used dot stickers, counting them aloud as she added them in each step. She then prompted the student to replicate this process for subsequent steps, while simultaneously encouraging him to record the results in the table. As the instruction progressed, there was a noticeable increase in the student's involvement. For example, for the task involving the function $y = n + 2$, and when asked about the general rule, he spontaneously expressed it through particularizations. Specifically, he answered the question involving the general term: 'well, [for position 100] it is 102 [points], because it is two more,' showing a functional correspondence relationship for the particular term 100, by adding 2 to it. Moreover, in the last session, in which the function involved was $y = n + 1$, he drew the first terms and from case 10 onwards he answered in a straightforward manner. For example, for case 25 he answered, '26. Because you skip one,' showing a functional relationship of correspondence by relating that the difference between the position and the number of points is 1, which led him to generalization, since he answered for the general term: 'You have to skip one.'

Student C maintained a good performance during the follow-up phase, with a mean of 94%, without overlapping with the baseline. In general, he showed no difficulties in the follow-up sessions, and made up for the lack of manipulatives by drawing the pattern.

For the Transfer sessions, he initially showed difficulties in understanding the situation of both tasks, but later he responded correctly by alluding to the context. For example, in the 'theme park' task, for case 10 attractions, he answered: '12. Because you have to pay also the thing here [pointing to the entrance] to enter the park.' And when asked about the general term he particularized with an example and answered: '102. Because you always give two [euros] more to enter the park,' showing a functional correspondence relationship. For the reverse case, when asked how many rides he could ride with 16 coins, he said: '14. When you go in you have to pay [the entrance].'

He maintained what he learned 5 weeks after the last training session, answering correctly to all the questions.

Internal validity

Inter-rater agreement was 94% during baseline, 98% during training, 86% in the follow-up, 100% in the Transfer to real-life context task, and 93% during maintenance. The mean interobserver reliability agreement for each student across all phases was 93% for student A, 93% for student B and 100% for student C. The procedural reliability was 93% for training and 100% for baseline, follow-up, transfer and maintenance. The mean procedural reliability was 91% for Student A and 100% for students B and C.

Social validity

In the initial interviews conducted before the start of the instruction, the students expressed their difficulties in solving math problems, which was also confirmed by their families. The teachers reported that their students struggled with understanding mathematical tasks expressed in verbal statements. However, in the post-instruction interviews, the students reported feeling comfortable during the intervention. The tutor of Student A stated that using manipulative materials such as dot stickers and sticks was particularly helpful in promoting abstraction skills. He also believed that this instructional approach could be applied to other mathematical topics and in regular classrooms to promote inclusion and benefit all students. The families appreciated the individualized support provided during the intervention and observed that their children's motivation increased as the study progressed.

Discussion and conclusions

The purpose of the present study was to examine the effects of explicit instruction through multiple representations with mediation to improve the performance of students with ASD on generalization tasks in functional contexts. A functional relationship was observed between multiple representation intervention and the improvement of the three participants' performance on generalization tasks. This work is a contribution to the field of interventions aimed at mathematical learning in children with autism (Polo-Blanco & González, 2021; Root et al., 2017), in particular, those aimed at fostering generalization as part of algebraic thinking (Goñi-Cervera et al., 2022).

In this work, generalization skills are developed in visual linear pattern tasks distinguishing between questions involving different terms. As shown in Table 5, students performed well on tasks involving consecutive terms (mean 92% success rate in the training evaluation versus 39% in the baseline), followed by those involving nonconsecutive terms (mean 92% success rate in the training evaluation versus 2% in the baseline), and showed more difficulty in tasks involving the general term (mean 46% success rate in the training evaluation versus 0% in the baseline). Other studies with typically developing students in similar grades also show that many were capable of generalizing (Pinto et al., 2022; Torres et al., 2022), and that those who encountered difficulties in generalization benefited from mediation (Narváez & Cañadas, 2023).

Given the scarcity of studies on early algebra in students with ASD, this work is a contribution to the way in which these students express generalizations in functional contexts. At the same time, given the age and educational level of the participating students, this study adds to the research currently being done in the field of early algebra. It is of interest to note the richness of performing this type of activities in elementary school students with ASD, despite their communication difficulties (APA, 2013). In fact, task comprehension was, in general, successful for questions involving particular terms. More complex were the questions about the general term and, in particular, with the meaning of the word 'any,' regardless of the function involved ($y = n + 1$, $y = n + 2$, $y = n + 3$ or $y = n + n$). This is in line with the previous study on generalization in students with ASD (Goñi-Cervera et al., 2022) that showed that most participants had difficulties expressing generalization in an abstract sense. However, the students in our study were able to express generalization by relying on concrete numbers that they considered 'big.' For example, when they said: '400 + 1,' '1000 + 2' we can infer that they had identified the general rule, although they did not correctly express the term 'any,' except in some cases where they relied on objects or characters (e.g., Student B, with 'wing +2').

The use of the multiple representations (materials and tables) and the guideline sheet helped students, and especially student C, to structure the steps necessary to solve the task. The use of manipulatives such as the dot stickers facilitated the understanding of the task questions, as it allowed reasoning with concrete real cases. On the other hand, expressing the information in tables facilitated the resolution of the task for the three students. Students B and C relied on using it to organize the first cases and were able to abandon it as the cases grew. However, Student A showed the need to use it in

all cases, even during the follow-up sessions when the table was not provided to him. In those cases, he drew his own tables to solve even very large nonconsecutive cases which led him to generate incorrect answers. The instruction built upon the students' previous strategies used in the baseline phase, guiding them to transition from their own drawings to other representations in order to identify the regularity in each task. As Carpenter et al. (1999) claim, it is crucial to pay attention to children's reasoning and build learning based on their prior knowledge.

During the instructional process, the guiding questions linked to the mediation process (refer to Table 3) proved to be effective in helping students better understand the proposed task and focus on the relevant aspects of the task. Additionally, similar to findings from previous research with typically developing students (Narváez & Cañadas, 2023, Pinto & Cañadas, 2021) these actions helped students identify regularities, even if they did not always generalize them correctly. Our results agree with those of other authors on the effectiveness of various mediation actions in promoting students' responses and aiding in the correction of errors or difficulties (Ureña et al., 2019).

In addition, this study shows how students transferred their learning to everyday situations, reaching a mean accuracy of 75% during the Transfer phase. In relation to the two proposed transfer tasks ('theme park' and 'game of chairs'), we highlight the differences between the first one, with a mean success rate of 66% and the second task, with a mean success rate of 83%. In the first situation, the task had an important verbal component and the visual support provided (see Figure 3, left) differed from the visual supports used during instruction. This may have led to more difficulties in understanding the situation. In particular, two students considered that the attractions presented in the drawing were the only attractions to be considered and did not generalize to there being 'other attractions' that were not shown. This interpretation could be related to the type of literal thinking characteristic of people with autism (King et al., 2016). The second transfer task, although contextualized in an everyday game situation, was presented with a visual support in the form of an increasing pattern similar to those worked on in the instruction (see Figure 3, right), which surely facilitated the understanding of the task.

Limitations and future research

Although this study provides evidence of functional relationship identification among the participants, further research is required to better understand how students with autism express generalization. Previous studies, such as Goñi-Cervera et al. (2022), suggest that language difficulties may prevent students with autism from explicitly expressing relationships between quantities. Additionally,Ingelin et al. (2021) suggest that communication skills can be a challenge for students with autism when performing mathematical tasks. Despite these challenges, the present study was able to observe functional correspondence relationships through natural language, even though the students did not always explicitly refer to the variables. For example, Student B affirmed 'because there is one more,' referring to the fact that there would be one more child than chairs, and Student C answered '102, because you always give two more to enter the park.' It is worth noting that in the latter response, the term 'any' is being particularized to the number 100, and that the subsequent phrase 'coins more than attractions you want to enter' is omitted. However, it is important to exercise caution when interpreting these results, as language expression difficulties may have prevented students from identifying more generalizations.

Moreover, having established a relatively low mastery criterion of 75% might have led to reduced levels of generalization (transfer to real-life contexts) and maintenance. Recent research has yielded empirical evidence suggesting that certain criteria for mastery may align with heightened levels of generalization and proficiency maintenance (Fuller & Fienup, 2018; K. K. Wong et al., 2022). Analogous prospective studies might establish more demanding mastery criteria and analyze the impact it has on the generalization and maintenance of the acquired algebraic skills. Future studies should also aim to expand the age range of the participants and investigate whether students in lower and upper primary grades, or in the first years of secondary school, exhibit functional relationships when solving other types

of tasks. As previous research with typically developing students suggests (Pinto et al., 2022), it would be beneficial to explore tasks specific to early algebra that involve generalization in numerical expressions.

Finally, it is crucial to explore alternative models that may offer a more comprehensive understanding of students' challenges in generalization. One notable model worth considering that shares similarities to our proposed approach is the Instructional Hierarchy, which suggests that learning progresses across four key stages: acquisition, fluency building, generalization, and adaptation (Haring & Eaton, 1978). Moreover, fluency in basic skills, characterized by a combination of speed and accuracy (Binder, 1996), can be a crucial factor in increasing a child's likelihood of generalization. Although the current study focused on evaluating acquisition rather than fluency, the observed generalization might be explained by the fact that students who are more fluent are more likely to engage in generalization (Binder, 1996). Focusing on this aspect could enhance the depth of similar studies by exploring whether a lack of fluency in core skills hinders students' ability to generalize in tasks involving algebraic concepts.

Implications for practice

Building on previous research conducted on typically developing students (Narváez & Cañadas, 2023), the present study highlights the effectiveness of instructor-student mediation in facilitating advanced generalization strategies. The research was conducted in a highly intensive series of sessions which may not be practical in a real-world educational setting. It would be important to delve deeper into the specific aspects that require adjustments to align with educational practices, all while considering the needs of the students and the dynamics of a typical general education classroom. For instance, the tasks remain consistent for all pupils (e.g., Phillips et al., 1991), but what necessitates adjustment is the method of fostering comprehension. In this study, diverse representations have been employed for this purpose. In a standard classroom setting, the use of these representations must be promoted by the teacher, while respecting the varying learning times. It is essential to recognize that, for students with special needs, these learning times may extend beyond those of their typically developing peers. In summary, the findings of this study could serve as a stepping stone toward promoting inclusion in regular classrooms, and it is imperative to continue exploring how such interventions can be adapted and extended beyond individualized instruction to foster collaboration and facilitate the integration of students with autism.

Disclosure statement

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Ethical approval

The study received ethical approval by the Clinical Research Ethics Committee of Cantabria, code 2020.252.

Informed consent statement

Informed consent was obtained from all subjects involved in the study.

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